

COURSE TITLE:

Artificial Intelligence

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Embedded Multispectral Plant Stress Differentiation System Using Edge AI

PROJECT OVERVIEW:

This project proposes the engineering and experimental evaluation of an ultra-low-cost, entirely offline, battery-operated physical leaf-clip prototype powered by an ESP32 microcontroller and a discrete narrow-band multispectral sensor.

The primary objective is not merely to build a generic plant health monitor, but to experimentally investigate whether high-dimensional spectral reflectance fingerprints captured at the leaf level can be utilized by on-device machine learning to accurately differentiate between three specific conditions:

- Healthy leaves
- Nitrogen-stressed leaves
- Water-stressed leaves

While existing low-cost systems typically focus on detecting whether a plant is generally stressed, this project investigates whether embedded AI can reliably differentiate the underlying physiological cause of that stress. By shifting the diagnostic focus from simple stress detection to cause-specific differentiation, this project aims to create a highly portable, actionable decision-support tool for resource-constrained agricultural environments.

RESEARCH FOUNDATION

1. Botero-Valencia et al. (2024) – "A Portable Tool for Spectral Analysis of Plant Leaves That Incorporates a Multichannel Detector to Enable Faster Data Capture"

The researchers built a 3D-printed, portable leaf-clip instrument incorporating two 9-band AS7341 multispectral sensors and an ESP32 microcontroller to measure light reflection and transmission properties. They utilized a Multi-Layer Perceptron (MLP) deployed via TinyML to refine real-time data acquisition and calibration. This work successfully proved that the ESP32 paired with the AS7341 is highly capable of reliable multichannel spectral sensing at the edge. However, their research focused on creating a general-purpose spectral measurement and calibration tool rather than training an applied agronomic model designed to classify specific physiological plant stress types.

2. Lopin et al. (2025) – "Evaluation of Low-Cost Multi-Spectral Sensors for Measuring Chlorophyll Levels Across Diverse Leaf Types"

This study evaluated the performance of three commercial multi-spectral sensors (AS7262, AS7263, and AS7265x) for non-destructive chlorophyll estimation across various leaf morphologies. They investigated the biological relationship between specific spectral reflectance bands and chlorophyll levels, which serve as a direct proxy for plant nitrogen status. The researchers achieved a highly significant result, obtaining an R^2 of 0.95–0.96 for smooth leaves using the AS7265x sensor. While this provides excellent biological validation for these sensors, their architecture remained tethered to a desktop PC using Partial Least Squares Regression, lacking an embedded edge AI implementation or any investigation into differentiating nitrogen stress from water stress.

3. Habibullah et al. (2020) – "Low-Cost Multispectral Sensor Array for Determining Leaf Nitrogen Status"

These researchers built a custom 12-band multispectral sensor array utilizing both visible and near-infrared (NIR) wavelengths. Using spectral data collected from canola plants in a controlled greenhouse, they applied a k-Nearest Neighbors (KNN) machine learning algorithm to classify different nitrogen fertilization levels. The system achieved an 88.4% classification accuracy in controlled environments. A key limitation relative to our proposed project is that they utilized a bulky, power-hungry Raspberry Pi 3 for processing, and their experimental scope was limited exclusively to nitrogen status without attempting to mathematically distinguish it from overlapping physiological conditions like water stress.

Comparison Table

Paper	What It Demonstrated	Limitation	What We Take From It	What We Investigate Differently
Botero-Valencia et al. (2024)	ESP32 can run TinyML for real-time multispectral (AS7341) data capture inside a leaf-clip.	Built a generic spectral measurement/calibration logger, not a stress classifier.	Validates our core hardware architecture (ESP32 + AS7341 + leaf clip).	We will train and deploy a localized agronomic model specifically to classify and differentiate stress causes.
Lopin et al. (2025)	The AS7265x sensor accurately tracks chlorophyll/nitrogen ($R^2 = 0.95\text{--}0.96$ for smooth leaves).	Relied on an Arduino tethered to a PC for statistical analysis; did not investigate water stress.	Biological proof that our selected low-cost sensors have the required optical resolution.	We move the processing to the edge (ESP32) and formulate the ML task as multi-class differentiation.
Habibullah et al. (2020)	KNN machine learning can classify leaf nitrogen status using 12 discrete spectral bands.	Used a bulky, power-hungry Raspberry Pi; focused only on nitrogen stress.	Algorithmic proof that discrete spectral bands can be statistically binned using ML.	We port the ML to an ultra-low-power microcontroller and introduce water stress differentiation.

RESEARCH GAP AND PROPOSED CONTRIBUTION

Established Scientific Foundation

Extensive literature in precision agriculture and computer science has already established that:

- Multispectral sensing can capture detailed plant physiological information.
- Spectral reflectance measurements strongly correlate with chlorophyll density and nitrogen status.
- Low-cost spectral sensors can be successfully integrated with embedded hardware.
- Machine learning models can accurately classify general plant conditions based on reflectance.

We are not claiming the discovery or initial implementation of these individual concepts.

Our Identified Research Question

Instead of repeating established binary stress detection, our primary research question is:

"Can a low-cost multispectral leaf-clip combined with embedded machine learning distinguish between different physiological stress conditions, particularly nitrogen stress versus water stress, rather than simply detecting whether a plant is stressed?"

The Distinction and Its Importance

- Existing/Basic Approach: "Is the plant healthy or stressed?"
- Our Proposed Investigation: "Is the plant healthy, nitrogen-stressed, or water-stressed?"

This distinction is critically important for agricultural viability. A plant may exhibit similar early-stage physiological decline whether it is starved of nutrients or deprived of water. Because different causes require entirely different agronomic interventions, simply detecting general "stress" does not provide the farmer with actionable data. By attempting to separate nutrient stress from water stress entirely on an edge device, we aim to transition low-cost agricultural technology from an observational monitor to a prescriptive diagnostic tool.

Our Proposed Contribution

Our specific contribution is the system-level application and investigation combining these technologies for cause-specific stress differentiation. The project will contribute:

- Low-Cost Physical Leaf-Clip: A mechanically robust, ambient-light-blocking enclosure ensuring consistent optical measurements in field environments.
- Multispectral Sensing over RGB: Utilizing discrete narrow bands (including red-edge and near-infrared) to detect microscopic physiological changes prior to visible chlorosis.
- Local/Edge AI Classification: Processing high-dimensional spectral arrays entirely offline using an ESP32 microcontroller, eliminating the need for cloud computing or PC tethering.
- Differentiation of Stress Causes: Experimentally evaluating whether heavily quantized, lightweight machine learning algorithms can reliably untangle the overlapping spectral signatures of nitrogen deficiency and drought stress.

PROJECT DESCRIPTION

In precision agriculture, timely and accurate intervention is critical to optimizing yield and preventing resource waste. Smallholder farmers currently rely heavily on visual crop inspection; however, by the time symptoms such as leaf yellowing (chlorosis) or wilting become visible to the human eye or standard RGB cameras, irreversible biochemical damage has already occurred. Furthermore, visual detection often fails to distinguish between the root causes of poor crop health. While existing high-end solutions, such as commercial SPAD meters and hyperspectral UAV mapping, can detect early-stage physiological stress with high accuracy, they cost thousands of dollars and require complex, cloud-based data processing, rendering them entirely inaccessible to small-scale farmers.

To bridge this gap, this project proposes the design and evaluation of an ultra-low-cost, entirely offline diagnostic prototype. The system utilizes a custom 3D-printed, ambient-light-blocking leaf clip equipped with a broad-spectrum LED and a discrete multispectral sensor (such as the AS7341 or AS7265x). Rather than relying on broad RGB bands, this sensor captures narrow, highly specific wavelengths of light—including crucial red-edge and near-infrared bands—that reflect the internal cellular structure and chlorophyll density of the plant tissue. The sensor is directly interfaced with an ESP32 microcontroller, forming a completely self-contained Edge AI architecture.

The core objective of this project is to investigate whether this low-cost hardware can do more than merely detect general plant deterioration. The system will be evaluated on its ability to differentiate between three specific target classes: Healthy, Nitrogen Stress, and Water Stress. Because nitrogen starvation and drought cause overlapping but distinct changes in leaf optical properties, a robust machine learning approach is required. To facilitate this, we will develop a localized, highly controlled dataset by cultivating a sample crop group under strict nutrient and watering protocols.

The collected spectral data will be used to train a lightweight machine learning classification model, such as Random Forest or a Support Vector Machine (SVM). Using TinyML frameworks, this model will be quantized, converted into highly efficient C++ code, and deployed directly into the flash memory of the ESP32. This enables the physical prototype to perform on-device mathematical inference in milliseconds. When clamped onto a leaf, the device will capture the multispectral fingerprint, process it through the embedded AI model, and output an immediate, actionable diagnosis (e.g., "Nitrogen Stress - 84%") onto an integrated OLED display. By synthesizing multispectral hardware with Edge AI, this project seeks to deliver a highly practical, cause-specific diagnostic tool that empowers agricultural workers with laboratory-grade insights entirely offline.

KEYWORDS:

Multispectral Sensing, Edge AI, TinyML, Plant Stress Detection, Nitrogen Stress, Water Stress, ESP32, Machine Learning, Precision Agriculture